Present value CHEAT SHEET

THE GOAL AND THE YARDSTICK

real assets (what the firm buys) · financial assets (what it sells)

opportunity cost of capital: the return investors give up, on investments of the same risk

invest \$1,000 → \$1,100 in a year: 10 % > 7 % given up, so invest

goal: maximise market value: every shareholder gains, whatever their timing (\$1,100 vs \$1,070 elsewhere)

FUTURE AND PRESENT VALUE

$FV = PV(1 + r)^t$: \$1,000 at 6 %, 3 years → \$1,191.02

$PV = \frac{C_t}{(1 + r)^t}$: \$1,000 in 3 years at 6 % → \$839.62

discount factor $DF_t = \frac{1}{(1 + r)^t}$: $DF_3 = 0.8396$ at 6 %

a dollar today is worth more than a dollar tomorrow

edges: $r = 0 \rightarrow DF = 1$; 30 years at 6 % → \$0.17

NET PRESENT VALUE

$NPV = C_0 + \Sigma \frac{C_t}{(1 + r)^t}$ (only add money at one date)

one year at 7 %: $-1{,}000 + 1{,}100/1.07 = +28.04$

as risky as 12 %: $-1{,}000 + 982.14 = -17.86$

$-1{,}000 \cdot 300 \cdot 400 \cdot 500$ at 8 %: PVs 277.78 · 342.94 · 396.92

NPV = +17.63 (the plain sum says +200: r = 0)

spreadsheet: $= C_0 + NPV(rate, C_1 \ldots C_n)$

NPV(8%, −1000, 300, 400, 500) = 16.32: one year too far

EDGE TESTS

perpetuity: r → 0, no fund is enough; 2.5 % → \$2M

annuity: r → 0 gives the sum (49,985 ≈ 50,000); t → ∞ gives C/r

annuity due: r = 0, timing is worth nothing

growing perpetuity: g = 0 → C/r; g → r, no limit (needs g < r)

growing annuity: g = 0 → annuity; $g = r \rightarrow t \cdot C_1/(1 + r)$

PERPETUITIES AND ANNUITIES

perpetuity $PV = \frac{C}{r}$ (spend only the interest)

\$50,000 a year at 5 % → \$1,000,000

annuity $PV = C\frac{1}{r} - C\frac{1}{r(1 + r)^t}$ (perpetuity now − later)

\$10,000 × 5 years at 6 %: 166,666.67 − 124,543.03 = 42,123.64

annuity due = annuity × (1 + r): 44,651.06

loan payment = PV ÷ annuity factor: 30,000/4.2124 = 7,121.89

growing perpetuity $PV = \frac{C_1}{r - g}$ (pay out what is left after growth)

\$40,000 next year, +2 %, at 6 % → \$1,000,000

growing annuity $PV = \frac{C_1}{r - g}[1 - (\frac{1 + g}{1 + r})^t]$

the same for 20 years → \$536,674.58

Present value CHEAT SHEET

QUOTED AND EFFECTIVE RATES

$\text{EAR} = (1 + \frac{\text{quoted}}{m})^m - 1$: 12 % monthly → 12.68 %

m = 1: 12 % · 2: 12.36 % · 4: 12.55 % · 365: 12.75 %

continuous: $\text{EAR} = e^r - 1 = 12.75\ \%$; $\text{FV} = \text{PV}\, e^{rt}$

$1,000 at 12 % for 3 years → $1,433.33

match the rate to the period: 9 % a year effective

→ $(1.09)^{1/12} - 1 = 0.7207\ \%$ a month, not 0.75 %

NOMINAL AND REAL

(1 + nominal) = (1 + real)(1 + inflation)

9 % and 3 %: real = 1.09/1.03 − 1 = 5.825 %

shortcut 9 − 3 = 6 %: misses real × inflation ≈ 0.175

real with real, nominal with nominal:

rent 10,000 · 10,300 · 10,609 at 9 % → 26,035.71

9,708.74 a year (real) at 5.825 % → 26,035.71

mixed (nominal at real rate) → 27,598.53, wrong

CASE: BANGALORE COLD CHAIN (FICTIONAL)

A $15M today · B $2M a year at the start, +3 % · C $170k a month

added up: B 22.93 · C 20.40 (meaningless: different dates)

B paid a year late: 14.41 < 15; as paid: × 1.09 = 15.71

real check: $2M flat at 5.825 % → 15.71

C at 0.7207 % a month → 13.62 (at 0.75 %: 13.42)

B wins by 0.71 while r < 10.27 % and inflation > 1.82 %

CHECK BEFORE YOU TRUST IT

before you trust a valuation:

1 are the cash flows at the dates I think? (draw the timeline)

2 is the rate for the same period, and the same risk?

3 are the flows and the rate both real, or both nominal?

4 bring everything to one date, then add

SPREADSHEET AND PYTHON

FV(rate, n, pmt, pv, type) · PV(rate, n, pmt, fv, type)

PMT(rate, n, pv, fv, type) · NPER(rate, pmt, pv, fv, type)

RATE(n, pmt, pv, fv, type) · NPV(rate, C1, …, Cn)

EFFECT(quoted, m) · NOMINAL(effective, m) · EXP · LN

type 0: end of period · 1: start (annuity due)

money paid out is negative: PMT returns −7,121.89

Python: sum(c / (1 + r) ** t for t, c in enumerate(flows))